

AANCHAL RATHI

aanchalrathiwork@gmail.com | +91-9817669951 | LinkedIn | GitHub | LeetCode | GFG

Computer Science undergraduate specializing in AI & ML with experience in full-stack development, recommendation systems, and ML-powered web applications. Skilled in Python, JavaScript, REST APIs, databases, and cloud technologies.

Education

- **VIT Bhopal University, Bhopal** Sep 2023 – Present
Bachelor of Technology in Computer Science (AI & ML Specialization)
CGPA: 8.95 / 10
- **DAV Public School** 2019 – 2023
12th Grade: 83.8% (2023)
10th Grade: 94.1% (2021)

Technical Skills

- **Languages:** C++, Java, Python, JavaScript, SQL
- **Web Development:** HTML, CSS, Flask, REST APIs, Frontend-Backend Integration
- **Machine Learning:** Scikit-learn, TF-IDF, Cosine Similarity, Pandas, NumPy, Matplotlib, OpenCV
- **Databases & Cloud:** AWS, Microsoft Azure (Certified), MongoDB Atlas, Firebase Authentication, MySQL
- **Developer Tools:** Git, GitHub, VS Code, Postman
- **Core Concepts:** Data Structures & Algorithms, OOP, API Integration, Recommendation Systems

Projects

Youth News Hub – Personalized News Recommendation Platform

March 2026 – May 2026

Tech Stack: Python, Flask, JavaScript, HTML, CSS, MongoDB Atlas, Firebase Authentication, TF-IDF

- Developed a full-stack personalized news aggregation platform integrating multiple news APIs with automatic fallback handling for resilient content delivery.
- Built a TF-IDF and cosine similarity based recommendation engine to dynamically rank and personalize 15+ news articles per request based on user reading behavior.
- Integrated 2 external news APIs with automatic fallback handling and backend caching to improve content reliability under API failures and rate limits.

Power Consumption Analysis using Machine Learning

May 2025 – Jul 2025

Tech Stack: Python, Pandas, Scikit-learn, Matplotlib, Flask, HTML, CSS

- Built an analytics platform using smart meter and weather data to study household energy usage.
- Cleaned and prepared UCI dataset; trained models to detect inefficiencies like standby drains and HVAC overuse.
- Tuned hyperparameters, evaluated using accuracy and F1-score, and deployed via Flask dashboard.

Crop Recommendation using Machine Learning

Feb 2025 – Apr 2025

Tech Stack: Python, Pandas, NumPy, Scikit-learn

- Developed a model using soil nutrients, temperature, humidity, pH, and rainfall data.
- Performed EDA, feature selection, and trained Decision Tree, KNN, Naive Bayes, and SVM models.
- Achieved 95% accuracy with Decision Tree; supports data-driven crop recommendations.

Achievements & Profiles

- Solved 500+ DSA problems across LeetCode and GeeksforGeeks.
- Microsoft Azure Data Fundamentals Certified.
- IBM Skills Build - Getting Started with AI Certified.
- Machine Learning with Python certified from Coursera.
- Maintained CGPA of 8.95.

Leadership & Extracurriculars

- Core team member of the University Dance Club; managed and organized multiple events. [2024-Present]
- Won the Inter-College women's cricket tournament from our university. [2024]